

Technology Stack

Date	15 March 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	2 Marks

Define Your Technology Stack:

A technology stack is the complete set of technologies used to build a project from the user interface to the backend, database, machine learning model, deployment, and version control.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Bootstrap	Used to create a responsive, interactive, and user-friendly web interface for entering applicant details and displaying prediction results.

2	Backend / Server-Side	Python, Flask	Flask provides a lightweight web framework that integrates seamlessly with the machine learning model and handles prediction requests efficiently.
3	Database / Data Storage	CSV Dataset, Pickle (.pkl) Model Files	The CSV dataset is used for training/testing, while the trained ML model is stored as a Pickle file for fast prediction without retraining
4	Cloud / Hosting / Deployment	GitHub, Render	GitHub is used for version control and collaboration, while Render hosts the Flask application online for easy access.

5	Version Control & CI/CD	Git, GitHub	Git tracks project changes, and GitHub enables team collaboration, version management, and repository hosting.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib, Joblib	Scikit-learn is used for model training and prediction, Pandas and NumPy for data preprocessing, Matplotlib for evaluation graphs, and Joblib for saving/loading the trained model.